

Luca Ciampaglia

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Objective Statement:

- 3 years of research experience in mechanical design of bioinspired robotics, with multiple preprint publications and a research thesis in mechanism design and testing
- Looking to apply my skills with mechanical and electrical systems, rapid prototyping and fabrication, system control and simulation
- I excel at handling leadership and managerial responsibilities with over 3 years leadership experience

Education:

2024-2026	Northeastern University MS: Robotics, Mechanical Engineering Graduation Date: 05/2026	2020-2024	Case Western Reserve University BSE: Electrical Engineering Graduation Date: 05/20/20
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Research and Work Experience:

2024 - 2026 Graduate Researcher (M.S.) *Silicon Synapse Laboratory, Northeastern University*

PI: Dr. Alireza Ramezani

- Mechanical design of dynamic flapping-wing robot Aerobat
- Regulation of flapping gait thrust via actuator design and placement in wing structure
- Piezoelectric and tendon-driven actuation of lightweight aerial robots
- 3D Print-In-Place rapid fabrication of compliant joints and computational structures
- Experimental validation and testing of integrated mechanisms in Aerobat
- Actuator analysis of Husky Beta quadrupedal robot to bridge sim-to-real gap

2023-2024 Undergraduate Research Assistant *Soft Machines and Electronics Lab, Case Western Reserve University*

PI: Dr. Changyong Cao

- Cephalopod bioinspired water jet propulsion via bistable shell and soft actuator driven motion
- Finite Element Analysis (FEA) of rigid and hyperelastic shell deformation for fluid dynamics
- Hybrid pneumatic-tendon driven soft continuum manipulator control and simulation
- Custom simulation in Python and PyBullet software for deformable and elastic material modeling
- PCB testing and pneumatic system assembly

2023 Undergraduate Thesis: Automatic Guitar Tuner

Advisor: Dr. David Kazdan

- Analog signal processing chain, analog-digital conversion circuit
- Digital filtering, frequency analysis, and sensor-motor feedback control
- Actuator selection, mechanical design and rapid prototyping

Skills:

SOLIDWORKS:	Dassault Systèmes - CSWA, PDM	Robotics:	ROS, GitHub
Programming:	C++, Python, MATLAB, Java	Modeling:	Blender, Fusion360
Simulation:	PyBullet, Simulink, Simscape, Gazebo	Microsoft:	VSCode, Excel, Word, PowerPoint

Leadership Experience:

2021-2024 Team Administration (Risk Manager, Vice President, President) *Case Aeronautics Team, Case Western Reserve University*

- Managed team attendance, communication with university staff, and COVID-19 policies
- Guided team general body and administration meetings, managed team leadership and engineering decisions
- Managed multidisciplinary team collaboration to design, manufacture, and test UAVs
- Wrote and edited technical documents on the design and manufacturing of team-built UAVs
- Supervised team during travel to AIAA Design/Build/Fly Competitions in Tucson, AZ and Wichita, KS
- Assisted with treasury duties and funding procurement

References:

Alireza Ramezani, PhD *Silicon Synapse Lab*

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Lab Website: siliconsynapse.sites.northeastern.edu

Jason Bradshaw

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Changyong Cao, PhD *Soft Machines and Electronics Lab*

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David Kazdan, MD, PhD

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